

A Survey on Service Migration in Mobile Edge Computing

Mobile edge computing technology reduce the network operational cost, improve quality of service and enables a scalable internet of things architecture. However we cannot ensure the service continuity so we need service migration. There are two concepts similar to service migration which are live migration for data centers and Handover in cellular networks. Three-Layer framework augmented service migration flow is used to optimize the downtime and total migration time. In addition Data transferring optimization is another technique of migration running services. Data transferring process can be optimized from reduce data size, pipelined stages, dynamic adaption, workload distribution and interactive transfer for aliveness. "Follow me cloud prototype", "MDP base service migration" and "Time window based service migration" some of existing strategies of service migration. Virtual machine, container and agent are three technologies for the hosting mobile application components. Virtual machine technology currently use in data center but container based migration is new and more efficient than VM. Agent can perform better in the dynamic and heterogeneous environment in Mobile edge computing.

Host	Disadvantage	Advantage
Virtual Machine	Slow boot and running, large data volume to store and transfer	High ioslabiltiy and security
Container	Bad cross-platform performance	Less data, high starting speed
Agent	Preliminary stage and no existing framework to use directly	Administrative convenience, small data to transfer, raid boot and running

There are some issue and challenges in service migration in MEC which are an efficient edge server selection algorithm should be choose and virtual resource allocation on edge servers should be optimize with having security like Block-Chain technology. Another challenge is AI based service migration decisions with mathematical model. Those need further investigation and need more research.

Reference

- [1] Shangguang Wang , Jinliang Xu , Ning Zhang , Yujiong Liu 18 April 2018 A Survey on Service Migrationin Mobile Edge Computing [online] available at <https://ieeexplore.ieee.org/document/8340768>